Ogbonna Arinze

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 8 years of product development experience building scalable, mobile-first web applications and backend services across healthcare, fintech, and civic tech. Expert in modern JavaScript/TypeScript, React component architecture, and NodeJS server-side development. Proven track record of designing REST APIs with JWT authentication, implementing real-time features using WebSockets and SSE, and deploying cloud-native applications via Docker and AWS. Deeply experienced in integrating AI/LLM features, including RAG pipelines and multi-agent systems, to deliver intelligent, user-centric healthcare and data platforms.

CORE COMPETENCIES

- **Languages & Frameworks:** JavaScript (ES6+), TypeScript, React, Next.js, NodeJS, Express.js, NestJS, Vue.js.
- **Backend & Databases:** REST API Design, MongoDB, PostgreSQL, Prisma, Redis concepts, JWT/OAuth Authentication.
- **AI & Real-Time:** OpenAI API, LangChain, Vercel AI SDK, Mastra, RAG pipelines, WebSockets, SSE streaming.
- **Cloud & DevOps:** Docker, AWS (S3, CloudFront), Heroku, CI/CD pipelines, GitHub Actions, Nx monorepos.
- **Testing & Quality:** Jest, Enzyme, Playwright, Cypress, ESLint, Prettier, TDD/BDD, Lighthouse profiling.

PROFESSIONAL EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer (React / AI Integration) *May 2023 - Present*

- Spearheaded the redesign of a critical clinical trial data product, migrating from a Vue.js web application to a **React Chrome extension**, optimizing workflows for clinical site integrations.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma**, enabling users to query their data in real time via a conversational interface.
- Implemented secure **JWT-based authentication** and authorization workflows on the frontend, handling access and refresh tokens with HTTP-only cookies and enforcing role-based access control in the UI.
- Architected a reusable design system in **React** and Radix UI, improving UI consistency, accessibility, and developer efficiency across products.
- Optimized application performance by implementing lazy loading, code splitting, and image optimization, significantly reducing load times as measured by Lighthouse and Chrome DevTools.
- Increased stakeholder confidence by writing end-to-end tests with **Playwright** in the **CI/CD pipeline**, publishing test results on GitHub Pages using Allure Reports.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive user interface using **React**, achieving a 25% boost in user engagement and collaborating with architects to accelerate development by 30%.
- Enhanced the development workflow by introducing automated testing frameworks, including **Jest** and **Enzyme**, which led to a 20% decrease in bug reports.
- Optimized frontend performance and load efficiency by identifying render bottlenecks using React Profiler and Lighthouse Audits, reducing bundle size by 30-40% and achieving a 35% faster load time.
- Established engineering standards for code quality, testing, and deployment across the team using ESLint, Prettier, Husky, Lint-staged, and **GitHub Actions**.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, focusing on testable, maintainable, and well-documented server-side code.
- Translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team.
- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on Heroku and production servers on **AWS**.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by 80%.

- Enhanced system performance and user experience by adopting the native browser **EventSource API** for **real-time data streaming**, replacing conventional socket.io methods for faster data updates.
- Developed critical payment modules using **React** with Redux for state management, RxJS for data streaming, and Axios for asynchronous API requests.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side web application logic in **NodeJS** for multiple client-facing applications.
- Worked with a team of 5 to develop SureEdu, a school management portal for school owners to manage staff attendance, payroll, and student reports.
- Tested user-defined functions, triggers, and **REST API** endpoints to ensure robust backend integration with the frontend.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered platform for Nigerian budget analysis and corruption tracking.

- Architected and built the entire platform as an Nx monorepo managing 6 applications, including a **NestJS** (NodeJS) API backend, a **Next.js** citizen-facing web app, and a NestJS document ingestion pipeline.
- Designed and implemented a **multi-agent AI system** using the Mastra framework with 5 specialized agents, powered by a hybrid RAG pipeline combining pgvector similarity search, BM25 full-text search, and Cohere reranking.
- Implemented **SSE streaming** for **real-time chat responses** with keepalive heartbeats every 15s to prevent reverse proxy timeouts, alongside token-budgeted context management.
- Configured CI/CD with GitHub Actions for automated builds, Playwright end-to-end tests, and **Docker Compose** production deployment with multi-stage Dockerfiles and health checks.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*